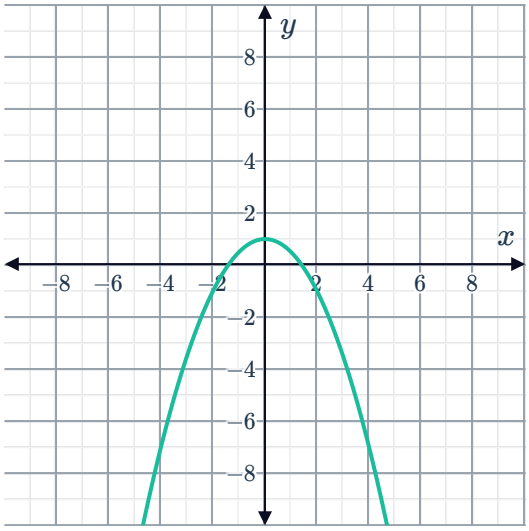
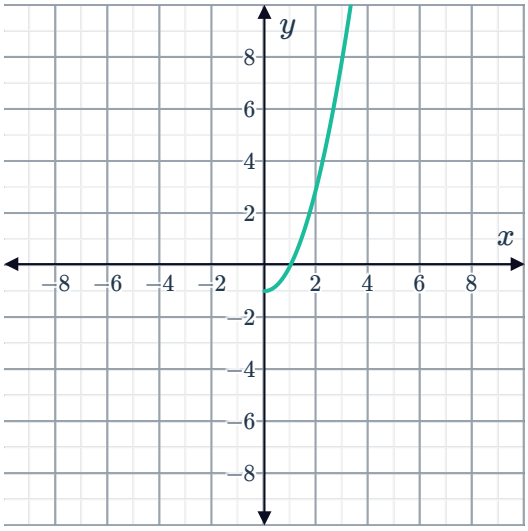
Does an inverse exist?

1 For each function drawn below, determine if it has an inverse:

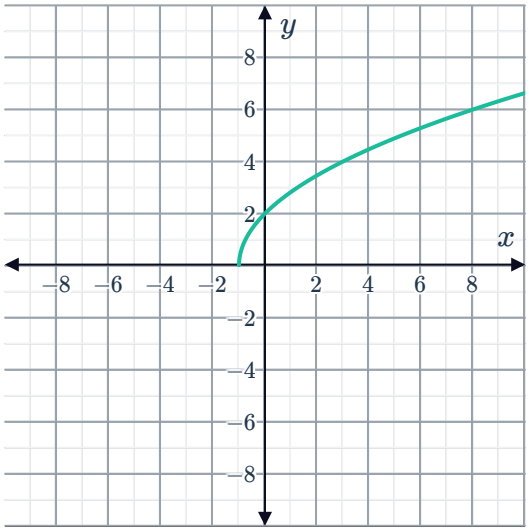
a



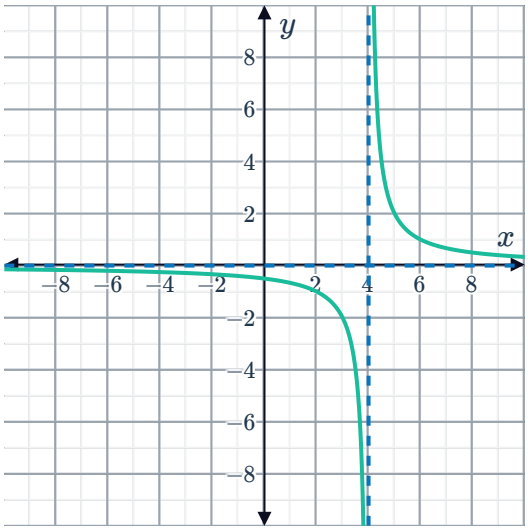
b



c

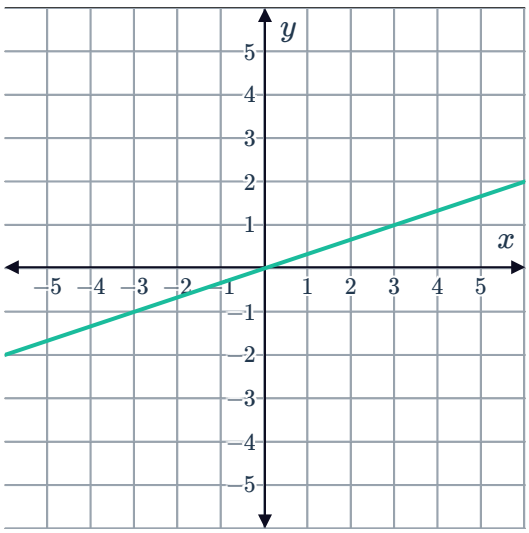


d

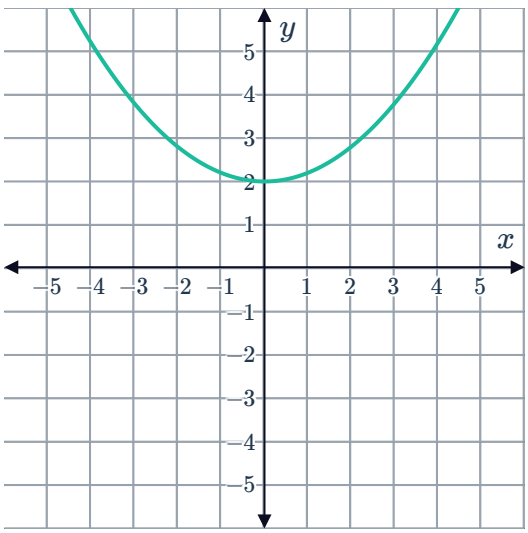


2 State whether the following functions have an inverse function:

a

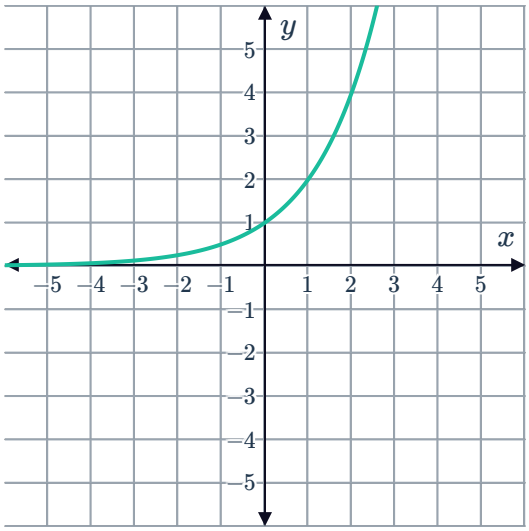


b

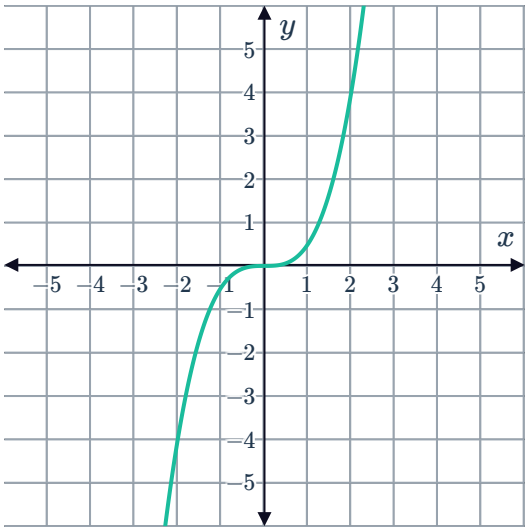


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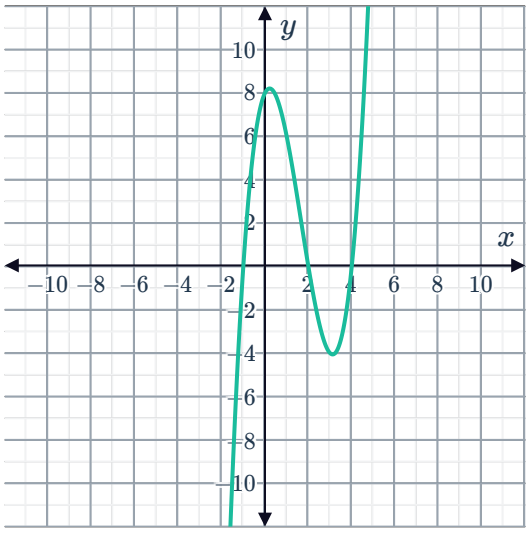
c



d



e



3 Determine whether the following functions have an inverse function:

a $\frac{7}{x}$ **b** $\sqrt{7-x^2}$ **c** $\sqrt{x}+7$ **d** $1-7x$

e 7^x **f** $4x^3+8$ **g** $\sqrt{4+x}$ **h** $-\frac{1}{7}x^2$

4 Consider the function $f(x)=\frac{4}{x+4}$.

a Sketch the graph of the function.

b Is the function a one-to-one function?

c Find an expression for the inverse function $f^{-1}(x)$.

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5 For each of the following functions:

i Sketch the graph of $f(x)$.

ii State whether the function $f(x)$ is one-to-one.

iii State whether an inverse function exists.

a $f(x) = x + 5$

b $f(x) = \dfrac{2x+3}{2}$

c $f(x) = (x-2)(x+3)$

d $f(x) = x^2 - 1$

e $f(x) = (x-5)^3$

f $f(x) = \dfrac{1}{8-x}$ for $x \neq 8$

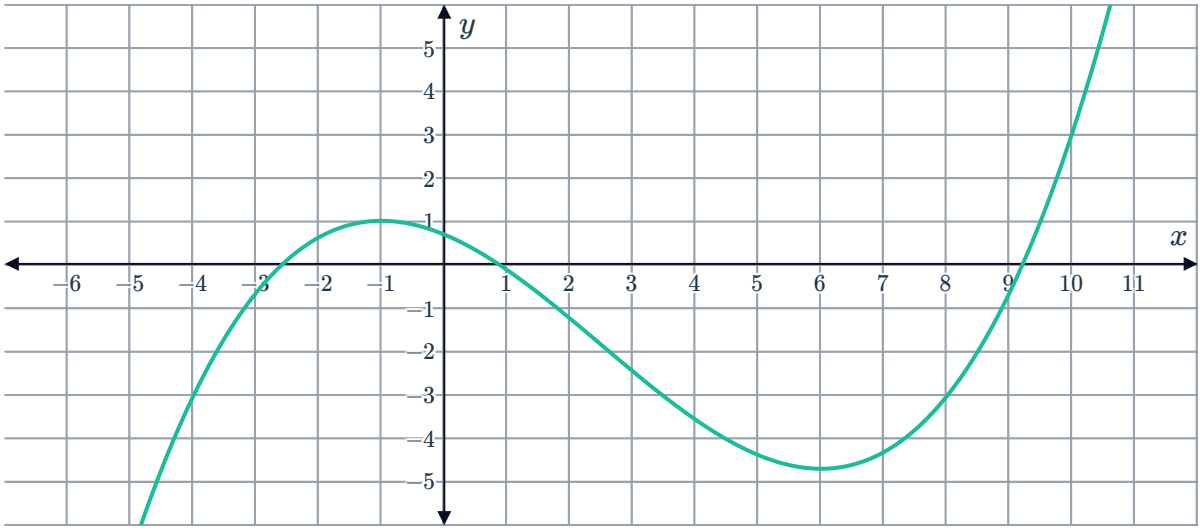
Restricting the domain

6 For many functions, we can determine an inverse relation by first breaking up the original function into parts that are one-to-one, and then finding the inverse of each part separately.

a Complete the function below to break $f(x) = (x-2)^2 - 6$ into two one-to-one functions that have the same rule as $f(x)$.

$$f(x) = \begin{cases} (x-2)^2 - 6, & x < \square \\ (x-2)^2 - 6, & x \geq \square \end{cases}$$

b Consider the graph of $y = g(x)$ shown:



Complete the function below to break the graph of $y = g(x)$ into three one-to-one functions that have the same rule as $g(x)$.

$$f(x) = \begin{cases} g(x), & x \leq \square \\ g(x), & \square < x < \square \\ g(x), & x \geq \square \end{cases}$$

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7 Find an appropriate restricted domain for the function $f(x) = (x-7)^2 + 12$ to have an inverse.

8 Consider the function $f(x) = (x-4)^2 - 6$ on the restricted domain $[4, \infty)$.

- **a** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain.
- **b** State the domain and range of $f^{-1}(x)$.

9 Consider the function $f(x) = (x+2)^2 - 9$.

- **a** State the domain restriction corresponding to the right half of the parabola described by $f(x)$, such that the restricted parabola is a one-to-one function.
- **b** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
- **c** State the domain and range of $f^{-1}(x)$.

10 Consider the function $f(x) = (x+4)^4$.

- **a** State the domain restriction corresponding to the right half of the quartic defined by $f(x)$, such that the restricted quartic is a one-to-one function.
- **b** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
- **c** State the domain and range of $f^{-1}(x)$.

11 Consider the function $f(x) = -(x-5)^2$.

- **a** State the domain restriction corresponding to the left half of the parabola defined by $f(x)$, such that the restricted parabola is a one-to-one function.
- **b** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (a).
- **c** State the domain and range of $f^{-1}(x)$.

12 Consider the function $f(x) = x^2 + 10x + 23$.

- **a** Complete the square for $f(x)$ to write the function in turning point form.
- **b** State the domain restriction that defines the right half of this function, making it one-to-one.
- **c** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (b).
- **d** State the domain and range of $f^{-1}(x)$.

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13 Consider the function $f(x) = x^2 + 4x + 3$.

- **a** Factorise $f(x)$ fully.
- **b** Find the axis of symmetry of $f(x)$.
- **c** State the domain restriction that defines the right half of this function, making it one-to-one.
- **d** Find the inverse function $f^{-1}(x)$ for $f(x)$ on the restricted domain found in part (c).
- **e** State the domain and range of $f^{-1}(x)$.

14 The inverse of a function $f(x)$ is $f^{-1}(x) = \sqrt{8 - x}$.

- **a** State the domain and range of $f^{-1}(x)$.
- **b** If $f^{-1}(x)$ is the inverse of $f(x)$, find $f(x)$.
- **c** State the domain restriction on $f(x)$.
- **d** What do you notice about the domain and range of $f(x)$ and $f^{-1}(x)$?

15 Consider the function $f(x) = \sqrt{x - 5} + 8$.

- **a** State the domain and range of $f(x)$.
- **b** Is the function $f(x)$ a one-to-one function?
- **c** Find the inverse function $f^{-1}(x)$.
- **d** Given that $f(x)$ is only half a parabola, what must be the domain restriction on $f^{-1}(x)$?
- **e** What do you notice about the domain and range of $f(x)$ and $f^{-1}(x)$?

16 Consider the function $f(x) = (x - a)^4 + b$.

- **a** State the largest domain and range of $f(x)$ in terms of a and b that would make $f(x)$ invertible.
- **b** Find the inverse function $f^{-1}(x)$.
- **c** State the domain and range of $f^{-1}(x)$ in terms of a and b.

17 Consider the function $f(x) = -(x - a)^2 - a$.

- **a** What would be the domain restriction on $f(x)$ in terms of a if we only want to keep the right half of the parabola?
- **b** Find the inverse function $f^{-1}(x)$.
- **c** Determine the condition on the value of a that results in at least one intersection between the graph of $f(x)$ and the graph of $f^{-1}(x)$.

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18 The largest domain over which the function $f(x) = x^2 + bx + c$ has an inverse is $[3, \infty)$. The domain of the inverse function $f^{-1}(x)$ is $[-2, \infty)$.

a Find the value of b.

b Find the value of c.

Applications

19 An encryption tool takes in a 10-digit code, n, and puts it into the formula $K = (n - a)^4 + b$ to create an encrypted key, K. It is important that when we try to get the code back from the key there is only one possible value for n.
Determine whether the following restrictions should be imposed to ensure the correct code is returned:

a $K > b$ **b** $n > a$ **c** $n > b$ **d** $K > a$

20 The per capita CO_2 emissions of Norway have been recorded since 1950, and the tonnes per capita emissions M, at a time t years after 1950, can be approximated by the model $M = 2.95\sqrt{t} + 7.8$.

a Determine the per capita CO_2 emissions at the time when recording started.

b Express t as a function of M.

c Hence, find the number of years it took for the per capita emissions of 1950 to double. Round your answer to one decimal place.

21 The function $d(t) = 120 - 4.9t^2$ can be used to find the distance d that an object dropped from a height of 120 m has fallen after t seconds.

a Is the function one-to-one?

b Find the inverse function, $t(d)$.

c Prove that $d(t(d)) = d$.

d Prove that $t(d(t)) = t$.

e Are d and t inverse functions?

f Determine how long it will take an object to fall 41.6 m when dropped from a height of 120 m.

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